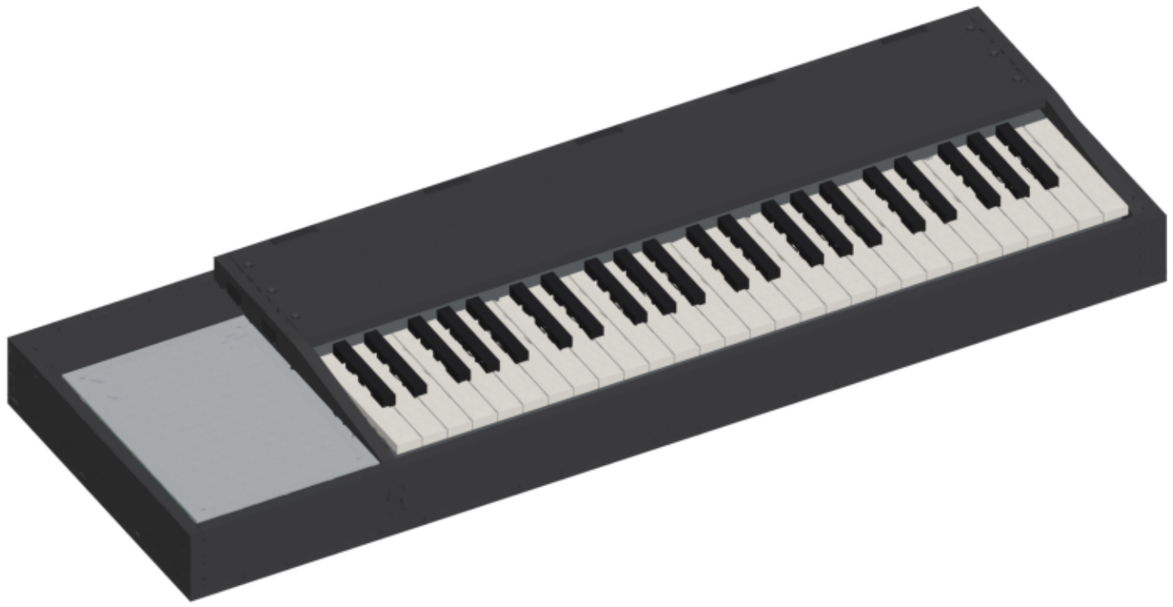
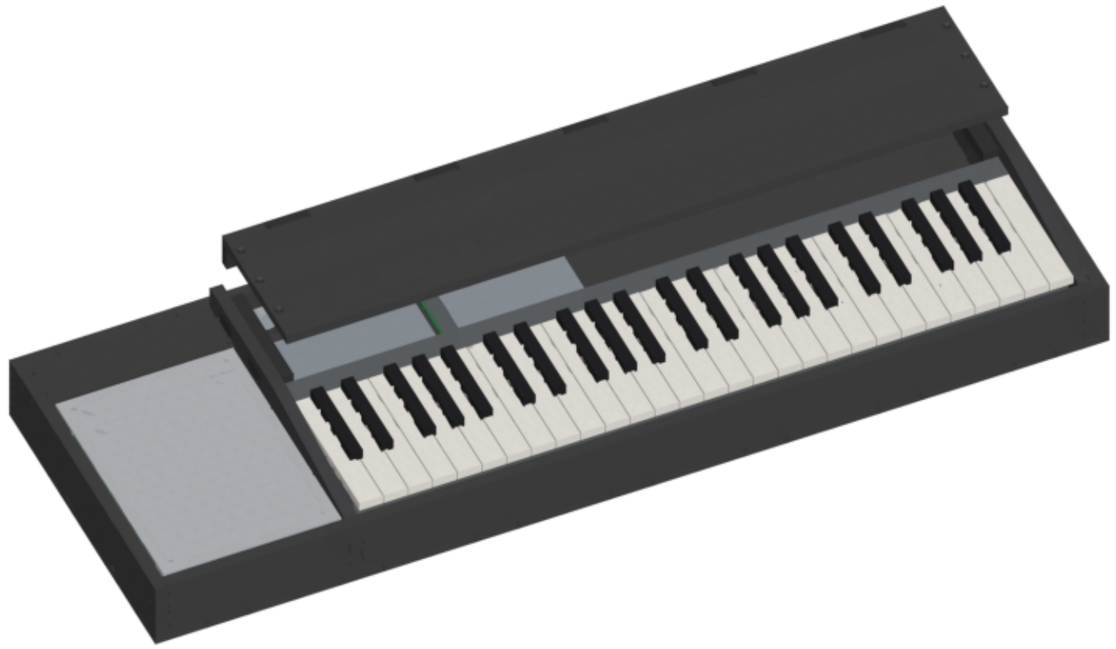


front left, strip in place



strip lifted: electronics bay (emulator, power supply)



rear: connector plate

Key data

Outside: 911 x 310 mm, height 67 (walls) / 90 (strip), plus feet 12

Birch plywood 12 / 9 mm, covered in black tolex, no lid

CNC-cut, plugged together: finger joints, tabs and slots, glued

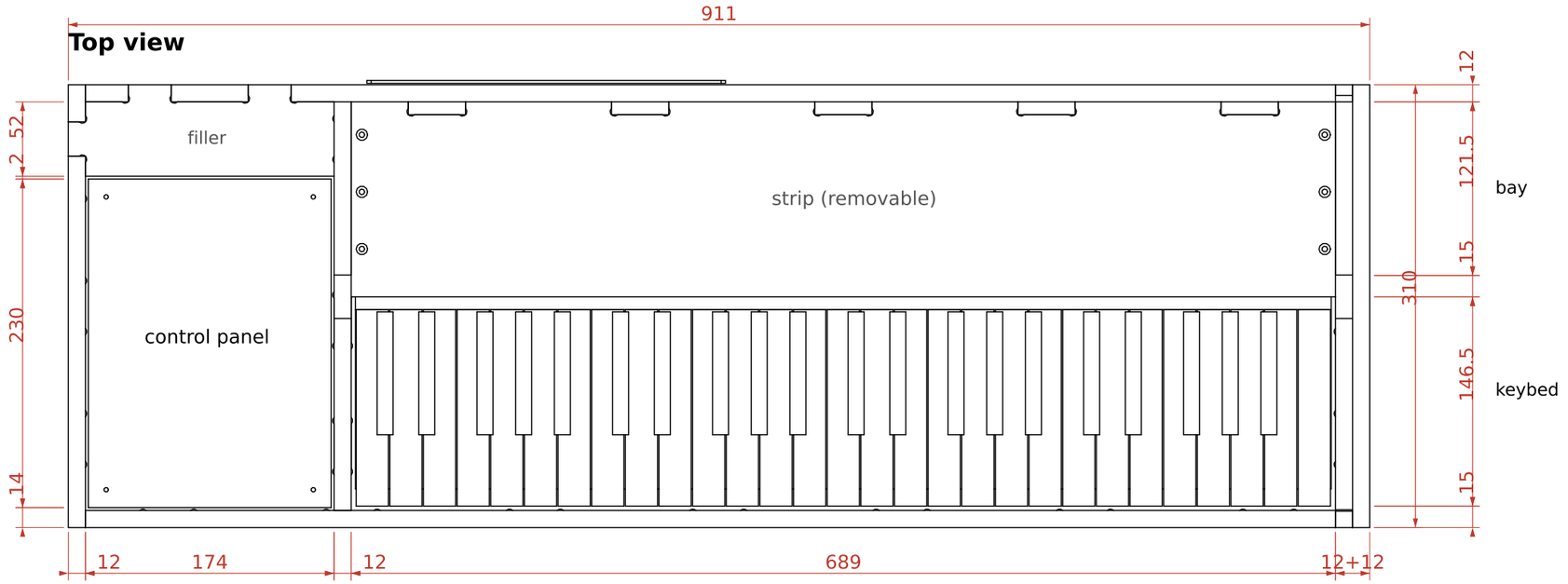
Keybed Fatar TP/9S 49 on two aluminium Z profiles 20 x 20 x 2, control panel 170 x 230 flush with the walls

Electronics bay under the removable strip: bus emulator 160 x 86, power supply 126 x 80

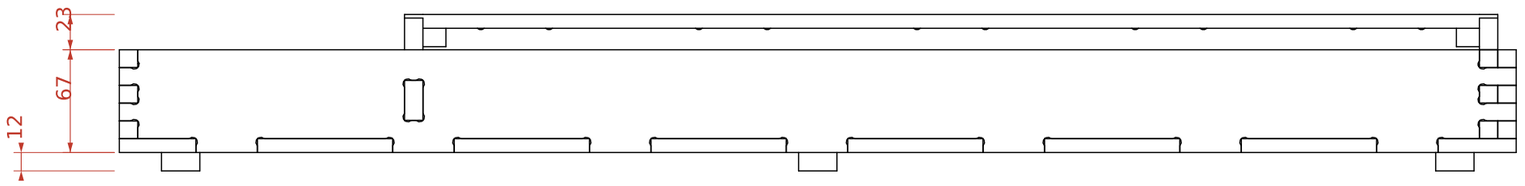
Rear plate (aluminium 2 mm): USB, DC IN, POWER, SUPPLY 2600/INT, MIDI IN, MIDI OUT, MIDI THRU

ARP 3620 replica - wooden case	Rev0.5 13.11.2026
Overview	sheet 1
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	renders not to scale

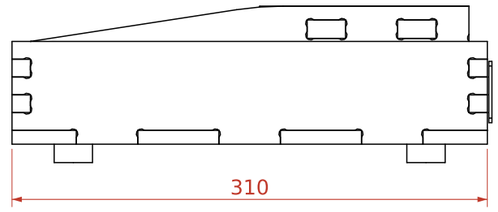
Top view



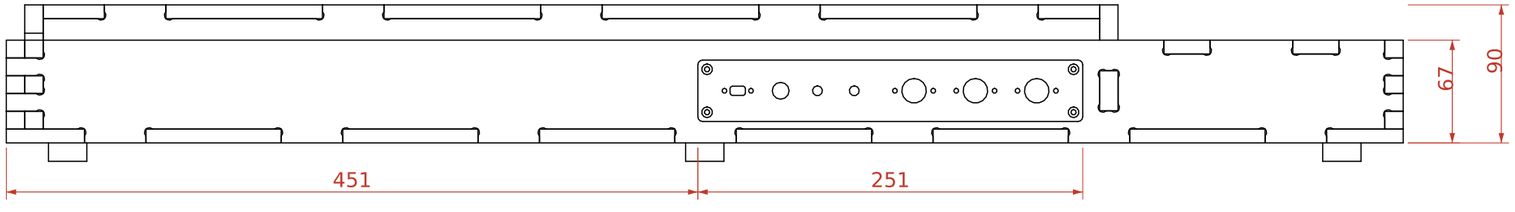
Front view



Right side

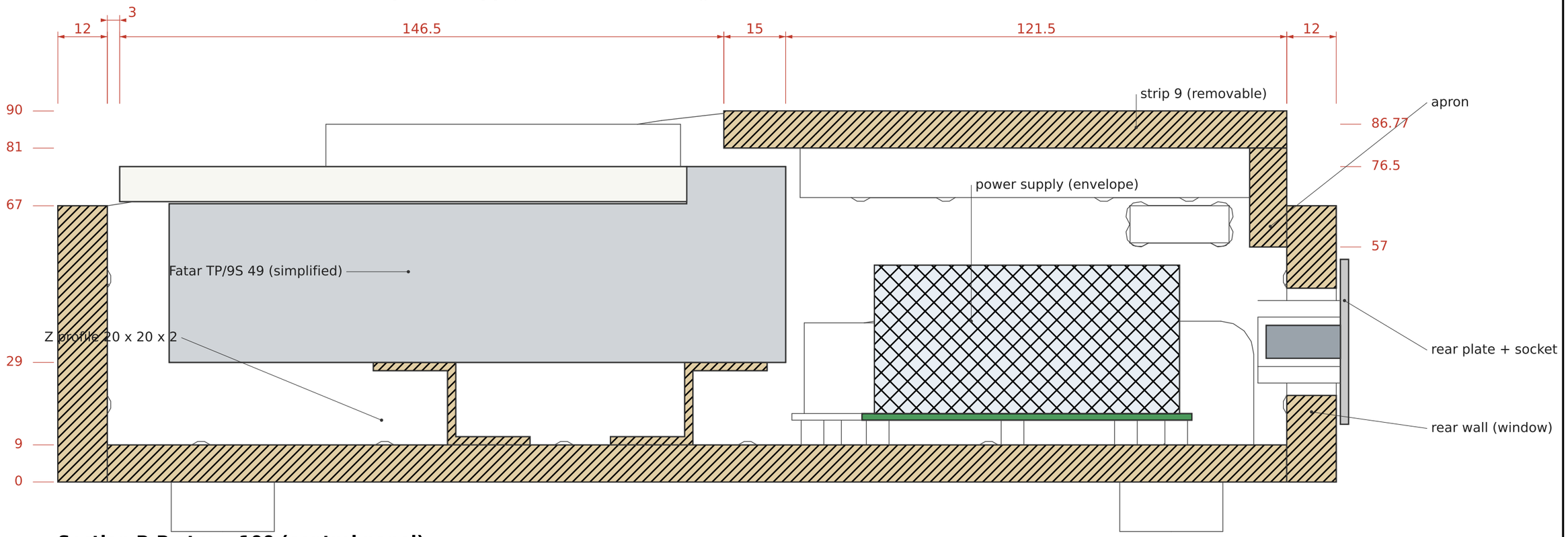


Rear view

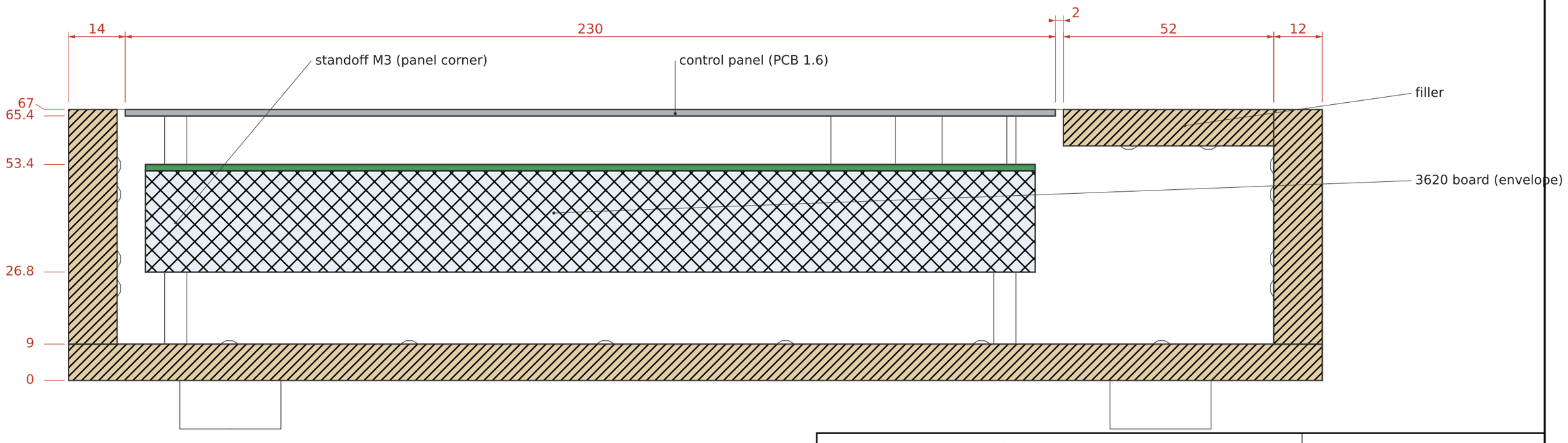


ARP 3620 replica - wooden case	Rev0.5 13.11.2026
General arrangement	sheet 2
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	scale 1:4

Section A-A at x = 440 (keyboard, power supply), seen from the right



Section B-B at x = 100 (control panel)

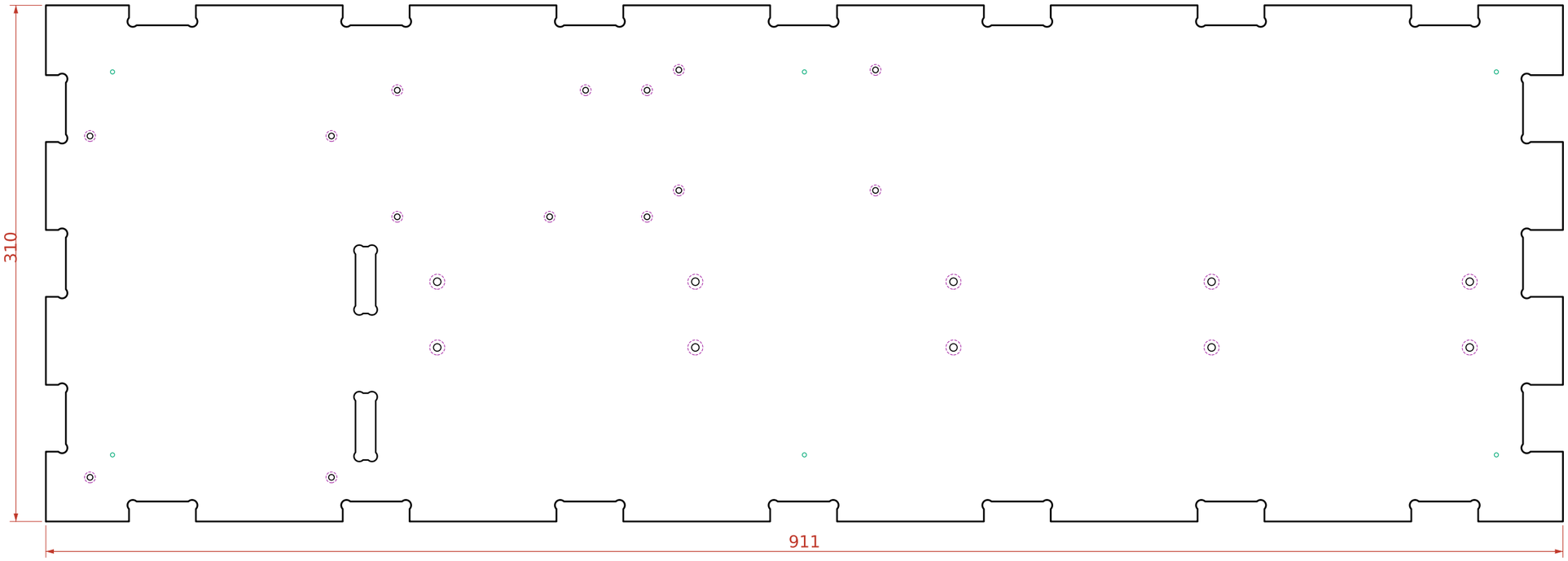


Heights follow from the keybed: the lower edge of the white key fronts (39 above the mounting plane, Fatar drawing 1:1, checked on the keybed) sits 1 above the wall top. The keybed stands on Z profiles 20 high - all heights follow from both (params.py: KB_LIP, T_RISER). Board spacer 12 and 25 for parts and plugs under the 3620 board are estimates. Electronics are simplified boxes (parts height emulator 22, power supply 36).

ARP 3620 replica - wooden case	Rev0.5 13.11.2026
Sections	sheet 3
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	scale 1:1

Base - 1 x

birch plywood 9 mm, seen from above [base]



seen from above (inside). Countersinks: dashed = on the underside. Coordinates x/y from the front left corner.

dia 4.5 csk 9 far face (10 x) Z profile M4 x 12
235/104.5 390/104.5 545/104.5 700/104.5 855/104.5 235/144 390/144 545/144 700/144 855/144

dia 3.4 csk 6.5 far face (4 x) panel standoff M3
26.5/26.5 171.5/26.5 26.5/231.5 171.5/231.5

dia 3.4 csk 6.5 far face (6 x) emulator standoff M3
211/183 302.52/183 361/183 211/259 324.11/259 361/259

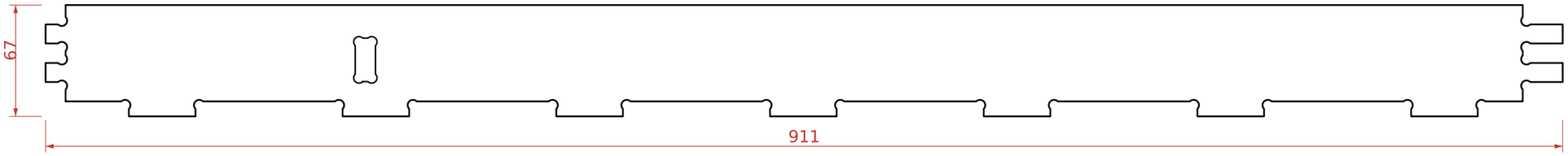
dia 3.4 csk 6.5 far face (4 x) power supply standoff M3
380.08/198.81 498.19/198.81 380.08/271.2 498.19/271.2

pilot 2.5 (6 x) foot
40/40 455.5/40 871/40 40/270 455.5/270 871/270

ARP 3620 replica - wooden case	Rev0.5 13.11.2026
Base	sheet 4
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	scale 1:3

Front wall - 1 x

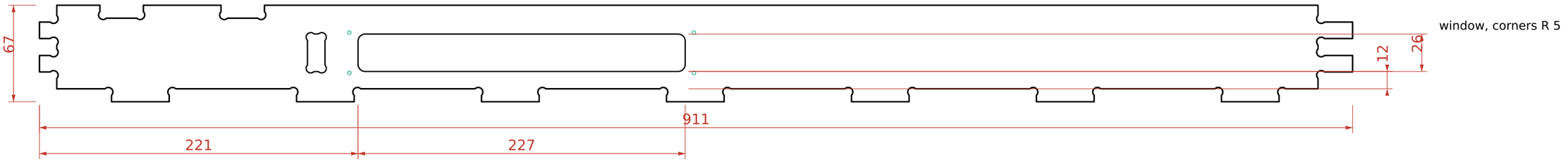
birch plywood 12 mm, seen from the front [front]



Seen from outside. Fingers at both ends, tabs into the base, slot for the left cheek. The right end takes end wall and right cheek together (24 deep).

Rear wall - 1 x

birch plywood 12 mm, seen from the front [rear]



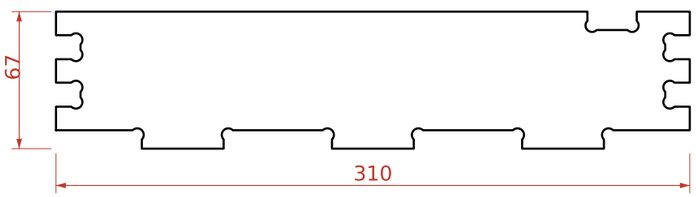
window, corners R 5

Seen from inside. Notches in the top edge for the filler tabs, slot for the left cheek, window for the bodies of the rear plate parts.

pilot 2.5 (4 x) rear plate, screw 3.5 x 16
215/20 454/20 215/48 454/48

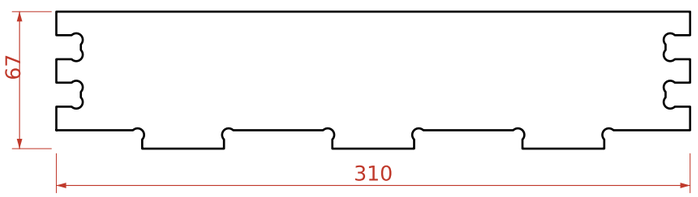
End wall left - 1 x

birch plywood 12 mm, seen from the right [end_l]



End wall right - 1 x

birch plywood 12 mm, seen from the right [end_r]

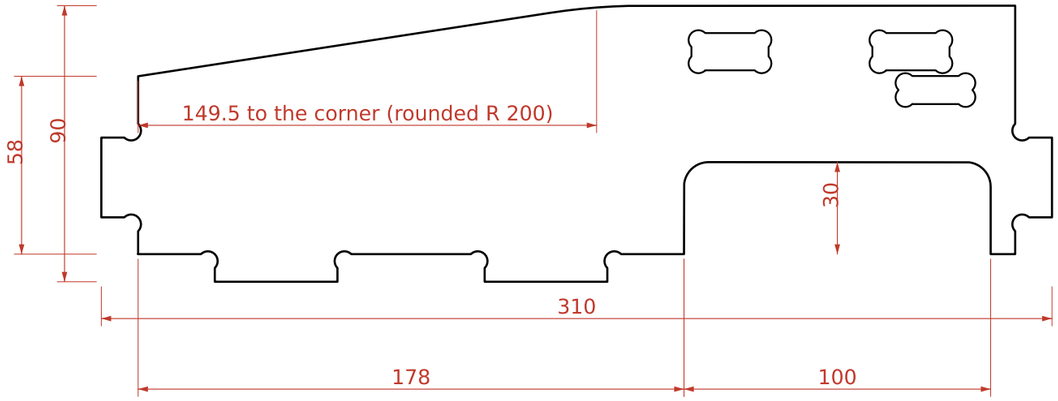


End walls seen from the right. Fingers at both ends (bottom, middle and top finger belong to the end wall), tabs into the base; the left end wall has a notch in its top edge for the filler tab. Heights: 0 = top of the base, the tabs reach 9 below it.

ARP 3620 replica - wooden case	Rev0.5 13.11.2026
Walls	sheet 5
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	scale 1:3

Cheek left - 1 x

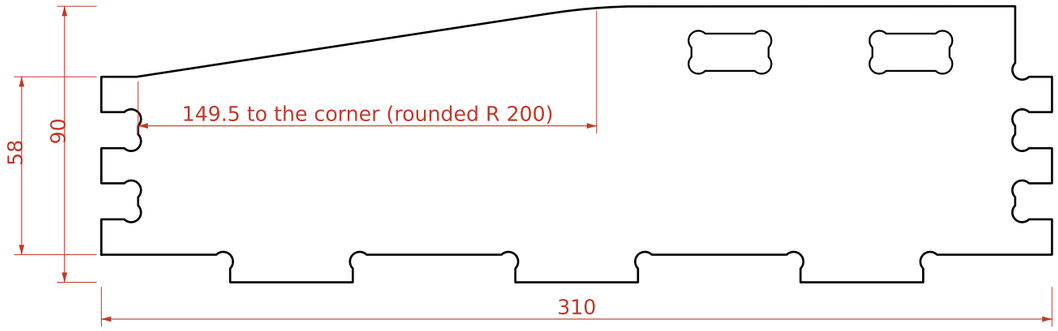
birch plywood 12 mm, seen from the right [cheek_l]



Seen from the right. u = 0 at the inner face of the front wall,
v = 0 at the top of the base; the tabs reach 9 below it.
Slope from 58 to 81 high, blending into the
flat top with R 200 (tangent points in the DXF).
Two slots for the strip cleats. Tolex on both faces and the top edge.
Tabs: front and rear through the walls, down through the base.
Slot at the rear for the filler tab.
Cable notch 100 x 30, corners R 8: cables
from the electronics bay to the panel section.

Cheek right - 1 x

birch plywood 12 mm, seen from the right [cheek_r]

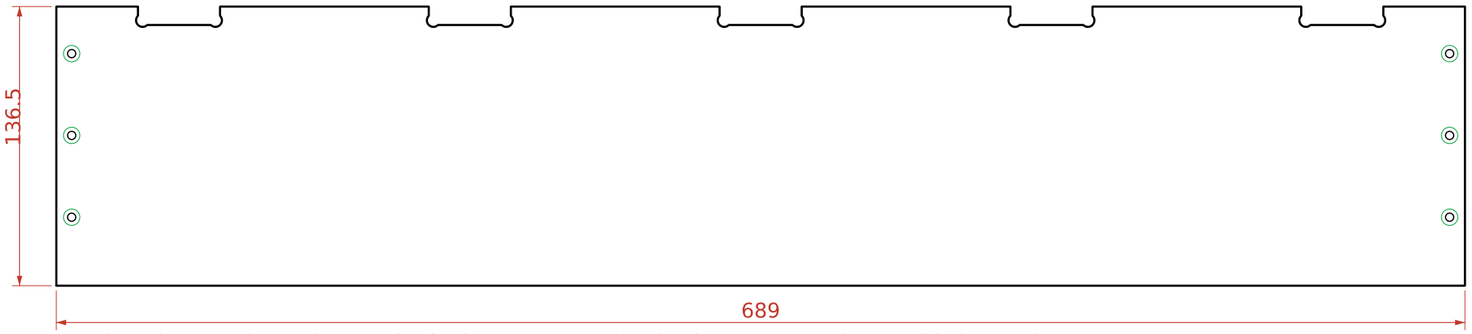


Seen from the right. u = 0 at the inner face of the front wall,
v = 0 at the top of the base; the tabs reach 9 below it.
Slope from 58 to 81 high, blending into the
flat top with R 200 (tangent points in the DXF).
Two slots for the strip cleats. Tolex on both faces and the top edge.
Stands against the right end wall and shares its joints:
same fingers into the front/rear wall, same tabs into the base.

ARP 3620 replica - wooden case	Rev0.5 13.11.2026
Cheeks	sheet 6
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Strip behind the keys - 1 x

birch plywood 9 mm, seen from above [strip]



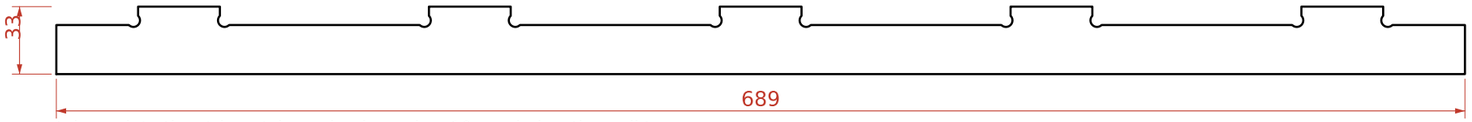
seen from above; notches at the rear edge for the apron; screws into the cleats, countersunk on top (black screws)

dia 4 csk 8 this face (6 x) screw 3.5 x 20 into cleat

7.5/33.5 681.5/33.5 7.5/73.5 681.5/73.5 7.5/113.5 681.5/113.5

Apron under the strip - 1 x

birch plywood 9 mm, seen from the front [apron]



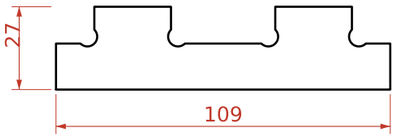
tabs up into the strip notches, glued; reaches 10 mm below the wall top

Keybed frame: two aluminium Z profiles 20 x 20 x 2, length 670 (bought, not CNC)

top flanges centred under the rows of keybed holes, bottom flanges point inwards; rivet nuts M4 in both flanges
keybed screwed onto the profiles from above, then the unit screwed to the base from below

Strip cleat - 2 x

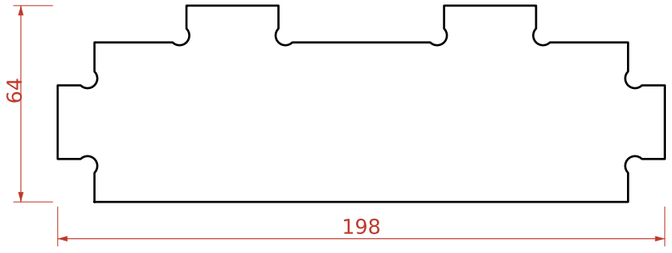
birch plywood 12 mm, seen from above [cleat_l, cleat_r]



lying flat, tabs through the cheek;
top 9 below the strip top

Filler behind the panel - 1 x

birch plywood 9 mm, seen from above [filler]



tabs into the left end wall, the rear wall and through the left cheek;
flush with the wall top

ARP 3620 replica - wooden case

Rev0.5 13.11.2026

Strip, cleats, filler, Z profiles

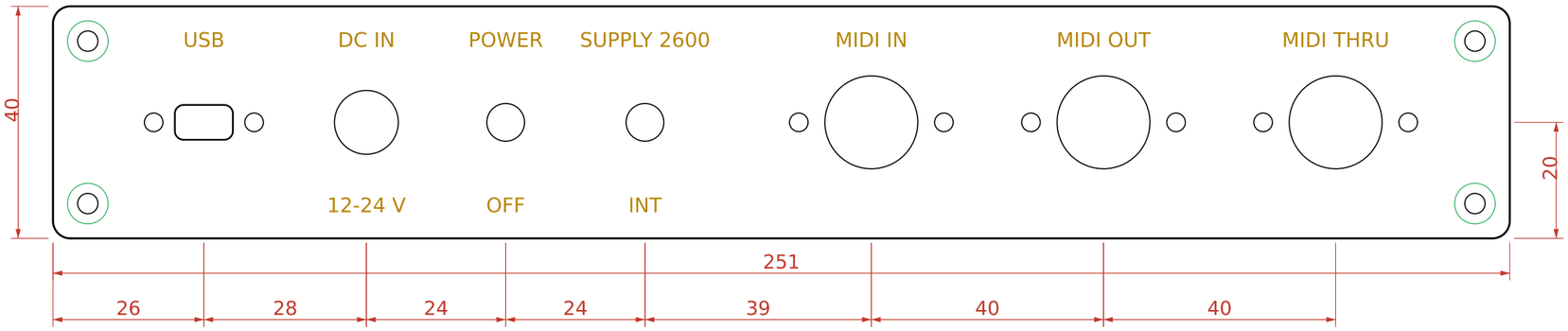
sheet 7

scale 1:3 / 1:2

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Rear plate - 1 x

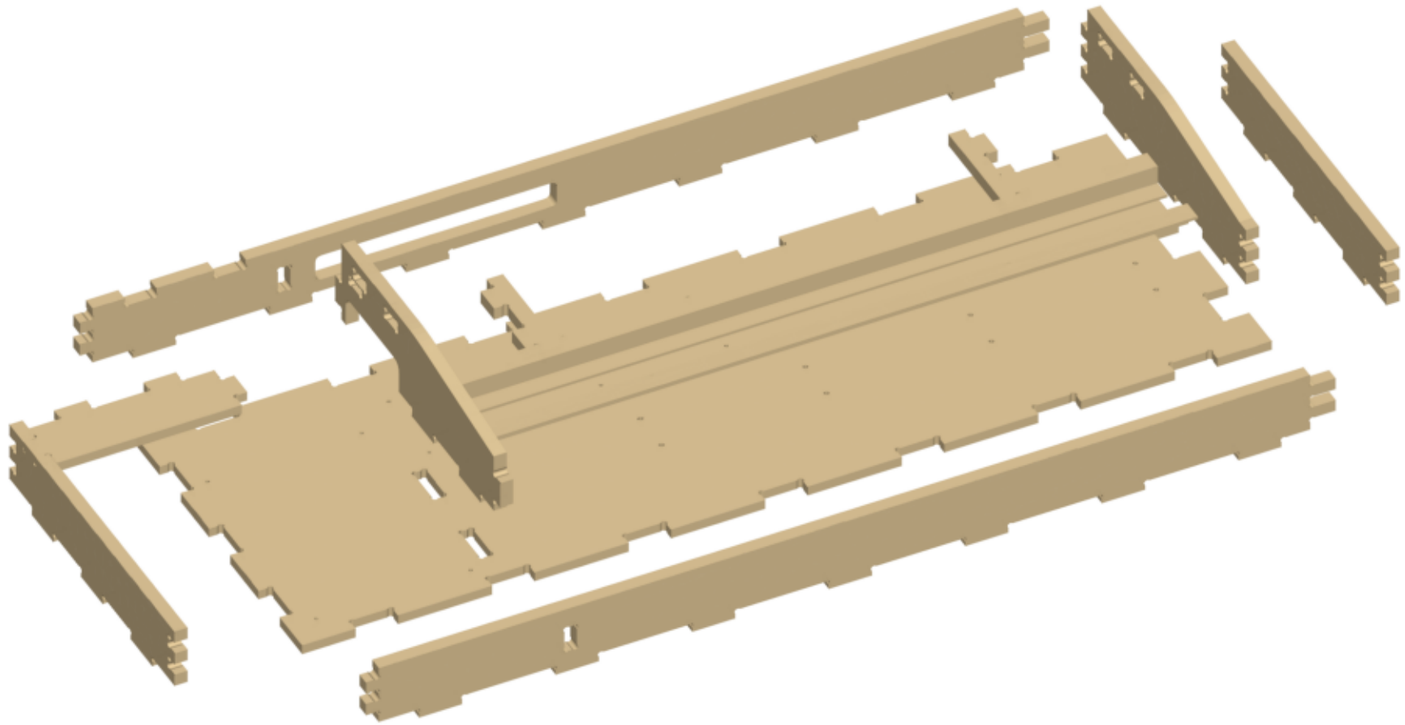
aluminium 2 mm, seen from the rear [plate]



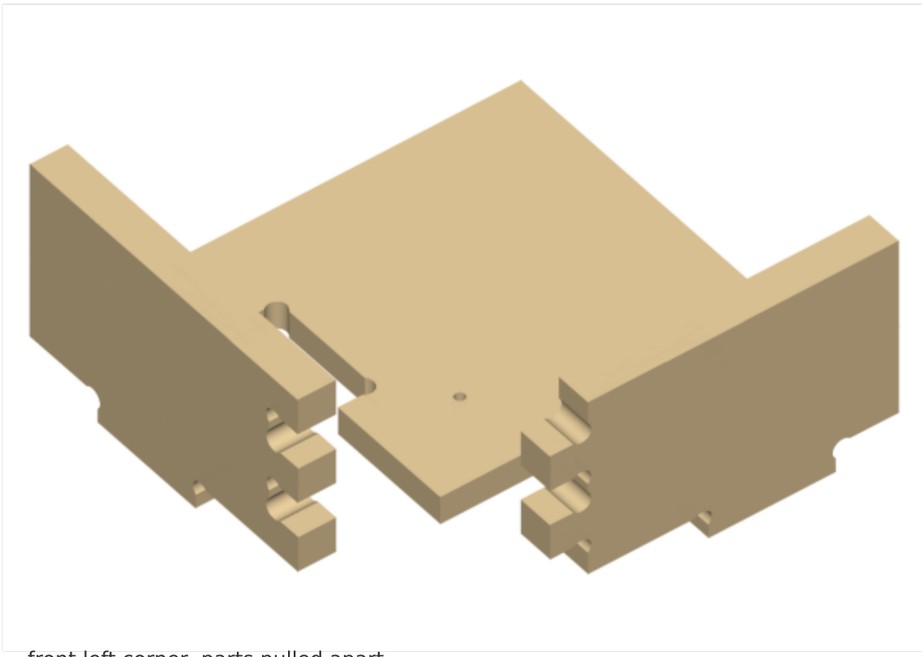
dia 16 (1 x) MIDI IN
 141/20
dia 16 (1 x) MIDI OUT
 181/20
dia 16 (1 x) MIDI THRU
 221/20
dia 11 (1 x) DC IN
 54/20
dia 6.5 (1 x) POWER
 78/20
dia 6.5 (1 x) SUPPLY 2600
 102/20
dia 3.5 csk 7 this face (4 x) screw 3.5 x 16
 6/6 245/6 6/34 245/34
dia 3.2 (2 x) USB
 17.35/20 34.65/20
dia 3.2 (2 x) MIDI IN
 128.5/20 153.5/20
dia 3.2 (2 x) MIDI OUT
 168.5/20 193.5/20
dia 3.2 (2 x) MIDI THRU
 208.5/20 233.5/20

Aluminium 2 mm, black anodised, legend engraved or printed white.
Seen from outside (from the rear of the case).
Mounted on the outside of the rear wall, over the tolex,
with 4 countersunk screws 3.5 x 16 (pilot holes in the rear wall).
Seen from the rear, the plate starts 451 from the left end of the case;
its lower edge is 14 above the underside of the base.
The window in the rear wall takes the bodies of the parts.
Behind the emulator 34 mm depth is free; the USB panel-mount
cable (Adafruit 3258) sits at the left end, behind the power supply.
Hole sizes are placeholders until the parts are chosen.

ARP 3620 replica - wooden case	Rev0.5 13.11.2026
Rear plate	sheet 8
CC BY-SA 4.0 (c) 2026 Dsl-man.de - github.com/dslmande/arp3620-woodcase	scale 1:1

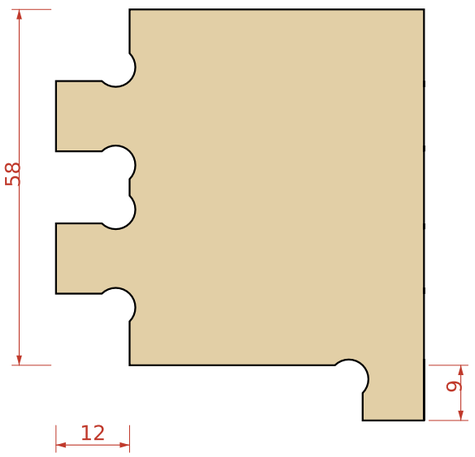


plugged together: finger joints, tabs and slots (exploded)

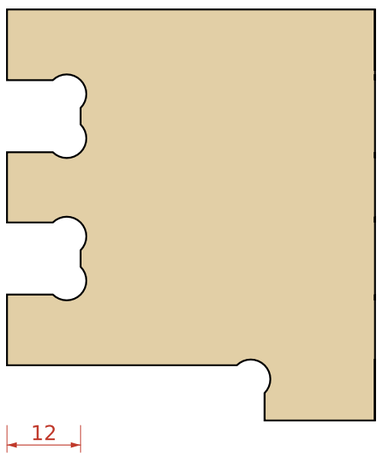


front left corner, parts pulled apart

front wall, left end (seen from outside)



left end wall, front end (seen from inside)



Dogbones R 3.2 in every inside corner (end mill up to 6.35 mm).
5 fingers over 58 mm: bottom, middle and top finger on the end wall.
Play per joint: notches and slots 0.15 mm wider than the tab
(split over both parts at the fingers). Joint depth = sheet thickness:
measure the plywood and enter it before cutting (params.py, T_*).
Tab ends sit flush with the outside, under the tolex.

Joints in the 3D model: no part overlaps another, every joint touches,
each part slides into place in the order below (model/check.txt).

Assembly order (carcass)

1	Base	
2	Keybed Z profile	down
3	Keybed Z profile	down
4	Cheek left	down
5	Cheek right	down
6	Strip cleat	to the left
7	Strip cleat	to the right
8	Filler behind the panel	to the right
9	Front wall	to the rear
10	Rear wall	to the front
11	End wall left	to the right
12	End wall right	to the left

ARP 3620 replica - wooden case

Rev0.5 13.11.2026

Joints, CNC, assembly order

sheet 9

detail 1:1, renders not to scale

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Cut list (finished sizes, mm)

part	qty	material	length	width	covering / finish
Base	1	birch plywood 9	911	310	underside painted black, edges covered by the tolex of the walls
Front wall	1	birch plywood 12	911	67	tolex outside, over the top edge, 20 mm down the inside
Rear wall	1	birch plywood 12	911	67	tolex outside, over the top edge, 20 mm down the inside
End wall left	1	birch plywood 12	310	67	tolex outside, over the top edge
End wall right	1	birch plywood 12	310	67	tolex outside, over the top edge
Cheek left	1	birch plywood 12	310	90	tolex on both faces and the top edge
Cheek right	1	birch plywood 12	310	90	tolex on both faces and the top edge
Strip behind the keys	1	birch plywood 9	689	136.5	tolex on top, front edge and ends
Apron under the strip	1	birch plywood 9	689	33	black paint, glued into the strip
Strip cleat	2	birch plywood 12	109	27	black paint
Filler behind the panel	1	birch plywood 9	198	64	tolex on top and front edge
Rear plate	1	aluminium 2	251	40	black anodised, engraved white (or printed)

net area: birch plywood 9 mm: 0.41 m2, birch plywood 12 mm: 0.23 m2, aluminium 2 mm: 0.01 m2

Hardware

- Tolex, black (vinyl case covering), approx. 0.6 m2 incl. 25 % waste
  outside, see covering notes
- contact cement for tolex, wood glue (PVAc), wood filler
  all joints glued; filler for dogbones and joint lines under the tolex
- 2 x aluminium Z profile 20 x 20 x 2, length 670
  keybed frame (not CNC)
- 10 x countersunk screw M4 x 12, 10 x rivet nut M4 in the bottom flanges
  base -> Z profiles (from below)
- 6 x countersunk wood screw 3.5 x 20, black
  strip -> cleats (removable strip)
- 4 x countersunk screw 3.5 x 16, black
  rear plate
- 10 x pan head screw M4 x 10 with washer, 10 x rivet nut M4 in the top flanges
  keybed -> Z profiles, from above through the keybed holes (take the keys over the holes off)
- 4 x M3 standoff, total length 56.4 mm (e.g. 40 + 8 and a washer)
  panel corners (panel PCB 1.6), M3 x 12 countersunk from below, M3 x 6 black from the top
- 10 x M3 standoff 6 mm, 10 x M3 x 12 countersunk
  emulator (6), power supply (4)
- 6 x rubber foot dia 25 x 12 with screw
- rear plate parts: 3 x DIN 5-pin panel socket, DC jack 5.5/2.1, 2 x toggle switch, USB panel-mount cable Adafruit 3258
  hole sizes on the plate are placeholders - confirm with the chosen parts
- optional: 8 x case corner, carrying handle

Assembly

1. CNC: cut all parts as per DXF, drill and countersink (CSK_BACK = countersink on the far face).
2. Screw the keybed onto the Z profiles first (keys over the holes off); fit the unit into the case dry.
3. Dry fit the whole case first (order on sheet 9), then take it apart and glue.
4. base on the bench, inside up
5. Z profiles (with the keybed on them): screw from below through the base
6. left cheek: tabs down into the slots of the base
7. right cheek: tabs down into the notches at the right edge of the base
8. strip cleats: from the inside through the cheeks
9. filler: right tab into the left cheek, hold it level
10. front wall: over the cheek tabs, tabs into the base
11. rear wall: over the cheek tabs and the filler tabs
12. end walls from the sides: fingers, base tabs, filler tab
13. Strip unit: apron tabs up into the strip, glued; the strip stays removable (screws into the cleats).
14. Fill dogbones and joint lines on the outside, sand flush, round all outside edges to about R2.
15. Paint the inside black, then tolex outside with contact cement; wrap 20 mm over the top edges.
16. Rear plate, feet; panel on its standoffs, electronics on the base, keybed unit screwed from below.

Open points

- Sheet thickness: measure the plywood, enter it in params.py (T_*) and regenerate.
- Keybed screws: check length against the keys (foot hole 4.5).
- Board spacer 12 and parts height under the 3620 board.
- Rear plate parts and their hole sizes.
- Optional handle and case corners.

ARP 3620 replica - wooden case	Rev0.5 13.11.2026
Cut list, hardware, assembly	sheet 10
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